

## EFFECT OF ACUTE DEHYDRATION ON BODY FLUID COMPARTMENTS IN THREE RODENT SPECIES, *RATTUS NORVEGICUS*, *ACOMYS CAHIRINUS* AND *MERIONES CRASSUS*

M. HOROWITZ and A. BORUT

Department of Zoology, The Hebrew University, Jerusalem, Israel

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**Abstract**—1. After acute dehydration *Rattus norvegicus*, *Acomys cahirinus* and *Meriones crassus* lost 18.6, 12.8 and 7 per cent of their body weight. The decrease in body water (g/100 g animal) was 21.0, 11.2 and 3.3 (mean figures).

2. The relative changes in plasma volume due to dehydration were 34, 4.9 and 11 per cent respectively. Extracellular fluid volume decreased by 32, 0 and 18 per cent.

3. T-1824 half-life in the blood system decreased during dehydration in *R. norvegicus* and *M. crassus*. In *A. cahirinus*  $T_{\frac{1}{2}}$  increased.

4. Sources of water loss during dehydration and possibility of capillary permeability changes are discussed.

### INTRODUCTION

UNDER exposure to high temperature mammals lose large quantities of water. It is generally accepted that the water source immediately available under such conditions is the blood fluid. Since the amount of water loss can very quickly exceed the plasma volume and the exchange of water between the fluid compartments of the body is rapid, the amount of water loss is distributed between these compartments. Results of several studies (Adolf, 1947; Macfarlane, 1964; Schmidt-Nielsen, 1964) indicate that each fluid compartment contributes differentially to the total amount of water loss and that this situation differs in various mammals. In desert-living mammals one of the most important features of physiological adaptation to dehydration is the conserving mechanism of blood plasma volume, as found, for example, in the camel (Schmidt-Nielsen, 1964). This mechanism allows the animal to suffer a substantial fluid loss without affecting the function of the blood system. However, few data have been published on the effects of dehydration on small mammals. Recently Kutcher (1967) found in the Mongolian gerbil a conservation mechanism of relative blood plasma volume. Unfortunately, with the method of dehydration used—water deprivation at room temperature—no changes were found in the amount of body water of the animals after the dehydration. This was also observed in the kangaroo rat (Schmidt-Nielsen *et al.*, 1948) after water deprivation at room temperature, essentially the same method. Since it is quite impossible to dehydrate desert mice merely by water deprivation

the technique of acute dehydration (Lewis *et al.*, 1960) seems to be more suitable for studies on these small mammals. In the present study the effects of acute dehydration on changes in the blood plasma volume and extracellular fluid volume were examined in three different rodent species, each living in a different habitat. In addition, in order to point out the various sources of evaporating water, total body water was measured. The possibility that the maintenance of a constant blood plasma volume is accompanied by a constant or decreasing rate of protein leakage from the blood system was also checked.

## MATERIALS AND METHODS

### *Animals*

*Rattus norvegicus*, *Acomys cahirinus* and *Meriones crassus* were chosen for this study. *R. norvegicus* lives in comparatively moist habitats. *A. cahirinus* and *M. crassus* are both desert mice. *A. cahirinus* inhabits rock crevices, with a much harsher microclimate than that of burrows dug in the soil in which *M. crassus* lives (Shkolnik, 1966). Because of this the latter is less exposed in nature to dehydration than *A. cahirinus*. Laboratory-reared *R. norvegicus* (Zabar race) albino strain were obtained from the animal house of the medical school of the Hebrew University, Jerusalem. Specimens of *A. cahirinus* (spiny mouse) were trapped in the Judean Desert, mainly in the surroundings of Ein-Gedi, and specimens of *M. crassus* (fat jird) were trapped in the southern part of the Negev.

All three species were kept routinely in the animal house of the zoology department. Only healthy animals and non-pregnant females were used in the experiments. The animals were kept in metal cages at  $25 \pm 3^\circ\text{C}$ . *R. norvegicus* was fed on Purina Chow and tap water. *A. cahirinus* and *M. crassus* were fed on barley and sunflower seeds. The spiny mice got in addition fresh carrot and alfalfa. The experimental animals were fasted 24 hr before the experiment. For experiments, specimens of *R. norvegicus* weighing 110–130 g (male), *A. cahirinus* weighing 35–45 g (male and female) and *M. crassus* weighing 70–114 g (male and female) were used. Two separate groups of animals were used for the experiments. The first group of animals was used for measurements of tritiated water (HTO) space. The second group of animals was used for measurements of blood plasma and extracellular fluid volumes. In each of the two groups some of the animals were measured as non-dehydrated specimens, the others were dehydrated first and then assayed.

### *Procedure*

(a) *Dehydration*. Each animal was kept in a separate wire cage. Acute dehydration was obtained by exposure of the animals to  $37 \pm 1^\circ\text{C}$ , 15–20% r.h. This method is a modification of the method proposed by Lewis *et al.* (1960). Their conditions ( $42^\circ\text{C}$ ) had been found to be too extreme for our animals; they were found to survive longer at  $37^\circ\text{C}$  and dehydrate more. The experimental animals were exposed to these conditions until they suffered the maximal weight loss from which they were able to recover when offered water or fresh vegetables. Those limits were determined in preliminary experiments and by individual observation of each animal during the real experiment. It was assumed that by this procedure any adaptation of survival value to the dehydrated animals would be most clearly displayed. The animals were weighed before, after and, if necessary, during the experiment.

(b) *Anaesthetics*. The animals were anaesthetized for the administration of tracers and for sampling in blood plasma volume and extracellular fluid volume measurements. Ether was used as an anaesthetic for the rats, while Equi-Thesin (Jensen-Salsbery Lab.), 3–4  $\mu\text{l/g}$  body wt, was used for *A. cahirinus* and *M. crassus*.

The use of two kinds of anaesthetics may cause some difficulty in the interpretation of results. However, according to the experience of previous workers, ether has no influence

on blood volume (Wang, 1959). Barbiturates also have probably no influence on blood volume (Goodman & Gilman, 1965), so that measurements may be compared.

(c) *HTO space*. The volume of body water of the experimental animals, determined as HTO space, was measured as follows: tritiated water (New England Nuclear Co.),  $1\mu\text{C}/100\text{ g}$  body wt, was administered into the stomach with a catheter. Blood samples were obtained from the tail at intervals of 1 hr during 7 hr and the serum was assayed for tritium activity, using a liquid scintillation method. The method of Richmond *et al.* (1962) was used to measure and calculate the HTO space. Ten  $\mu\text{l}$  plasma of each sample was mixed with 15 ml Bray solution and 4% Cab-O-Sil (Rapkin, 1963) and measured for radioactivity after standing 24 hr in the dark at  $-4^{\circ}\text{C}$ . HTO space was calculated by extrapolation to zero time after equilibration of HTO with the body water.

(d) *Blood plasma volume, extracellular fluid volume and rate of disappearance of T-1824*. Blood plasma volume (PV) was determined with T-1824 dye and the extracellular fluid volume (ECV) as the inulin space, with  $^{14}\text{C}$ -labelled inulin (New England Nuclear Co.). The substances (100 mg/ml T-1824, 15% inulin carrier with  $0.3\mu\text{C}/\text{ml}$  inulin  $^{14}\text{C}$ ) were introduced, usually simultaneously, into the circulatory system (approximately 0.5 ml solution per 100 g body wt). The solutions were injected into the tail vein of the rats and intracardially in the other species.

Preliminary experiments for measuring the inulin space were first done on nephrectomized animals. After comparison of the inulin space data in nephrectomized and non-nephrectomized animals, which were found to be similar in the case of *A. cahirinus*, it was decided to work on non-nephrectomized *A. cahirinus*. In the case of the two other species this could not be done because of the high clearance rates of inulin. Accordingly, both kidneys of each animal were ligated before the inulin injection. Essentially the same results were obtained by PV measurements in nephrectomized or non-nephrectomized animals of the three species. Blood samples, approximately  $40\mu\text{l}$ , were obtained from the tail at 30-min intervals over a period of at least 2 hr. After centrifugation,  $10\mu\text{l}$  of plasma were taken for each analysis. T-1824 concentrations were determined with a Beckman Spinco-151 microcolorimeter, after dilution in 1 ml of water. The specific activity of inulin was determined by a Baird Atomic gas flow counter, after drying the samples. PV and ECV were calculated by extrapolation to zero time. The rate of disappearance of T-1824 from the blood stream was calculated from the extrapolation curves. This rate gives information on the rate of blood protein leakage (or protein destruction) and on the permeability of capillaries, since T-1824 couples with blood albumin (Rawson, 1942).

## RESULTS

### (a) *Effect of dehydration on body weight and on volume of body water*

When comparing the response of the three species to the experimental conditions, we see that the rat has a relatively short survival time (8–10 hr) and loses weight rapidly, but it can survive large losses (14–22 per cent) of its body weight. *A. cahirinus* and *M. crassus* lose weight much more slowly, both at approximately the same rate, but the spiny mouse can survive losses of 11–14 per cent of its body weight, while *M. crassus* can survive no greater loss than 5–9 per cent of its normal body weight (Fig. 1).

The weight loss of *R. norvegicus* and *A. cahirinus* are more or less equal to the amount of water loss (Table 1). In *M. crassus* apparently only half of its weight loss is water loss. The HTO spaces of non-dehydrated animals and animals after dehydration are presented in Table 1. The decrease of body water in 100-g *R.*

TABLE 1—BODY WEIGHT AND AMOUNT OF BODY WATER (HTO SPACE) IN NON-DEHYDRATED *R. norvegicus*, *A. californicus*, AND *M. crassus* AND IN THOSE EXPOSED TO DEHYDRATING CONDITIONS ( $37 \pm 1^\circ\text{C}$ , 15–20% r.h.)

| Species                | HTO space                                        |                                                          |                                                      |                                            |
|------------------------|--------------------------------------------------|----------------------------------------------------------|------------------------------------------------------|--------------------------------------------|
|                        | Initial body weight (g)<br>[Mean $\pm$ S.D. (n)] | % of decrease in body weight after dehydration           |                                                      | Amount of H <sub>2</sub> O lost per 100 g* |
|                        |                                                  | % of non-dehydrated body weight<br>[Mean $\pm$ S.D. (n)] | % of dehydrated body weight<br>[Mean $\pm$ S.D. (n)] |                                            |
| <i>R. norvegicus</i>   | 100.0 $\pm$ 20.0 (10)                            | 18.6 $\pm$ 4.0 (5)                                       | 67.6 $\pm$ 3.2 (5)                                   | 21.0                                       |
| <i>A. californicus</i> | 42.0 $\pm$ 2.8 (10)                              | 12.4 $\pm$ 1.4 (5)                                       | 60.6 $\pm$ 6.8 (5)                                   | 11.2                                       |
| <i>M. crassus</i>      | 103.0 $\pm$ 14.0 (10)                            | 7.0 $\pm$ 2.0 (5)                                        | 56.6 $\pm$ 6.9 (5)                                   | 3.3                                        |

\* The amount of water lost was calculated from the mean figures as the difference in the amount of water in 100-g animal before dehydration and the reduced body weight after dehydration.

TABLE 2—BLOOD PLASMA VOLUMES (PV) AND EXTRACELLULAR FLUID VOLUMES (ECV) IN NON-DEHYDRATED *R. norvegicus*, *A. californicus* AND *M. crassus* AND IN THOSE EXPOSED TO DEHYDRATING CONDITIONS ( $37 \pm 1^\circ\text{C}$ , 15–20% r.h.)

| Species                | PV                                               |                                                          | ECV                                                  |                                                          | % of relative reduction after dehydration *          |        |
|------------------------|--------------------------------------------------|----------------------------------------------------------|------------------------------------------------------|----------------------------------------------------------|------------------------------------------------------|--------|
|                        | Initial body weight (g)<br>[Mean $\pm$ S.D. (n)] | % of non-dehydrated body weight<br>[Mean $\pm$ S.D. (n)] | % of dehydrated body weight<br>[Mean $\pm$ S.D. (n)] | % of non-dehydrated body weight<br>[Mean $\pm$ S.D. (n)] | % of dehydrated body weight<br>[Mean $\pm$ S.D. (n)] | PV ECV |
|                        |                                                  |                                                          |                                                      |                                                          |                                                      |        |
| <i>R. norvegicus</i>   | 113.0 $\pm$ 17.90 (12)                           | 7.67 $\pm$ 1.025 (6)                                     | 5.00 $\pm$ 0.908 (6)                                 | 22.57 $\pm$ 1.766 (6)                                    | 15.30 $\pm$ 2.547 (5)                                | 34 32  |
| <i>A. californicus</i> | 39.0 $\pm$ 19.65 (14)                            | 4.10 $\pm$ 0.162 (7)                                     | 3.90 $\pm$ 1.135 (7)                                 | 21.00 $\pm$ 1.732 (4)                                    | 21.00 $\pm$ 0.400 (4)                                | 4.9 0  |
| <i>M. crassus</i>      | 82.7 $\pm$ 16.25 (14)                            | 5.80 $\pm$ 1.004 (8)                                     | 5.20 $\pm$ 0.600 (6)                                 | 17.75 $\pm$ 0.500 (6)                                    | 14.60 $\pm$ 0.139 (6)                                | 11 18  |

\* Calculated from the mean figures.

*norvegicus* under dehydration is 21 g and of 100-g *A. cahirinus* and 100-g *M. crassus* 11.2 g and 3.3 g respectively.

(b) *Blood plasma and extracellular fluid volumes*

The effect of dehydration on PV and ECV was not the same in the three species (Table 2). The laboratory rat suffered the greatest loss.

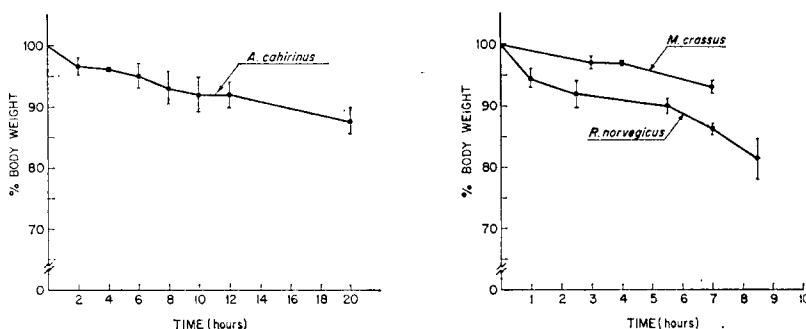


FIG. 1. Decrease in body weight of *R. norvegicus*, *A. cahirinus* and *M. crassus* exposed to dehydrating conditions of  $37 \pm 1^\circ\text{C}$ , 15–20% r.h. The data belong to the experimental group in Table 1. Vertical line: Mean  $\pm$  S.D. ( $n = 5$ ).

There was a significant ( $P \leq 0.01$ ) reduction of 34 per cent in PV and 32 per cent in ECV. *M. crassus*, too, suffered reduction in PV and ECV, being 11 per cent ( $0.1 < P < 0.45$ ) and 18 per cent ( $P < 0.01$ ) respectively. The picture is different for *A. cahirinus*: the changes in PV were too small to have any meaning and there were no differences in the ECV ( $P \geq 0.5$ ).

(c) *Disappearance of T-1824 from the blood*

Differences were found in the rate of disappearance of T-1824 dye from the blood after exposure to dehydration (Table 3). In rats and in *M. crassus* there is an increase in the rate of the dye disappearance. In *A. cahirinus*, however, the situation

TABLE 3—RATE OF T-1824 DYE DISAPPEARANCE FROM THE BLOOD PLASMA OF NON-DEHYDRATED ANIMALS AND THOSE EXPOSED TO DEHYDRATING CONDITIONS ( $37 \pm 1^\circ\text{C}$ , 15–20% r.h.) EXPRESSED AS THE HALF-LIFE ( $T_{\frac{1}{2}}$ ) OF THE DYE IN THE BLOOD SYSTEM

| Species*             | Non-dehydrated<br>$T_{\frac{1}{2}}$ (hr)<br>[Mean $\pm$ S.D. (n)] | Dehydrated<br>$T_{\frac{1}{2}}$ (hr)<br>[Mean $\pm$ S.D. (n)] |
|----------------------|-------------------------------------------------------------------|---------------------------------------------------------------|
| <i>R. norvegicus</i> | 50.0 $\pm$ 2.5 (6)                                                | 16.6 $\pm$ 6.6 (6)                                            |
| <i>A. cahirinus</i>  | 5.0 $\pm$ 1.4 (7)                                                 | 22.0 $\pm$ 20.5 (7)                                           |
| <i>M. crassus</i>    | 12.0 $\pm$ 4.8 (8)                                                | 7.7 $\pm$ 4.6 (6)                                             |

\* Weight data as in Table 2 (same experimental group).

is the reverse; there is no change or an increase in the  $T_{1824}$  half-life after the dehydration. (Part of the specimens show a significant increase of  $T_{\frac{1}{2}}$  while in others  $T_{\frac{1}{2}}$  are similar or little longer than that of the non-dehydrated).

### DISCUSSION

Figure 2 sums up the experimental data on the changes of body fluids under acute dehydration treatment. In the laboratory rat, which suffers the largest reductions in the shortest time, half of the evaporated fluids are from extracellular compartments (blood plasma and extravascular volumes). Presumably 50 per cent of the decrease is from intracellular sources. *A. cahirinus* shows another pattern of

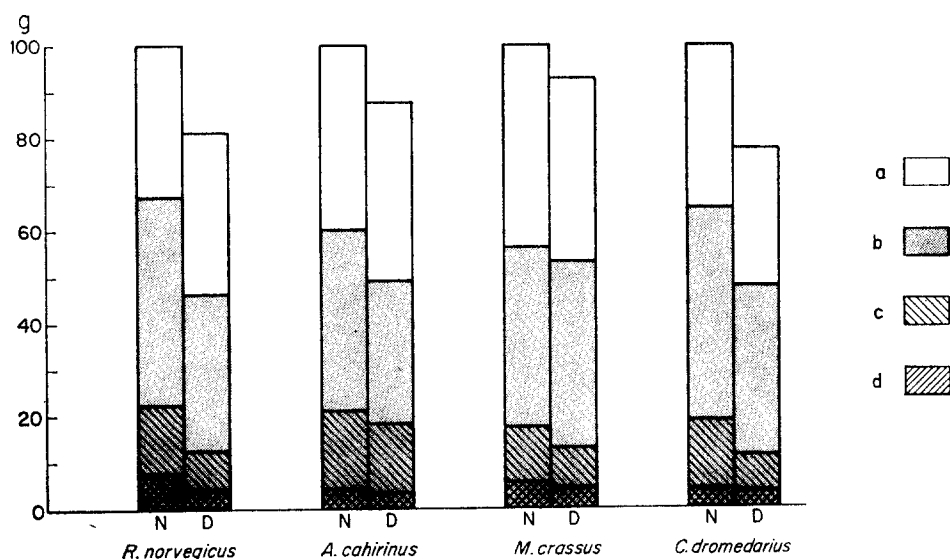


FIG. 2. Body fluid compartments in non-dehydrated and dehydrated *R. norvegicus*, *A. cahirinus*, *M. crassus* and *C. dromedarius* (from Schmidt-Nielsen, 1964) calculated as g/100 g initial body wt of non-dehydrated animal. Data on rodents obtained by pooling the results of Tables 1 and 2. a, Body weight; b, body water; c, ECV; d, PV; N—non-dehydrated animal; D—dehydrated animal.

water loss. Here the extracellular source contributes only 24 per cent of the total fluid reduction. The main part lost is probably from cellular sources. In *M. crassus* the ECV reduction is equal to the total water loss. In both *A. cahirinus* and *M. crassus* PV suffers small or no reduction during the dehydration treatment. *A. cahirinus* shows the same ability as the Saharan camel to conserve plasma volume while suffering dehydration. The water loss is mostly from compartments other than the ECV, differing in this respect from the camel (Fig. 2). In *M. crassus* the small reduction is correlated to the small amount of dehydration. Most of the loss is from extracellular sources. As only half of the weight loss (3.3 g out of

7.0 g) of the animal is accounted by water loss, there may be destruction of body compounds which slow down the dehydration. The same results were obtained when water content was calculated by method of direct weighing (Magazanik, 1967). We cannot explain this satisfactorily at the moment.

Analysis of the T-1824 half-life measurement ( $T_{\frac{1}{2}}$ ) furnishes some information on the plasma volume regulation in *A. cahirinus*, suggesting little seepage or destruction of proteins from the blood stream while undergoing acute dehydration. Such a mechanism may be instrumental in stabilizing the relative blood volume. The role of proteins in this mechanism was mentioned already by Schmidt-Nielsen (1964) who found that in the camel the half-life of T-1824 dye is long (2–4 weeks) as compared to its half-life in the dog (6–7 hr; Gregersen & Rawson, 1959) and in man (12 hr; Gregersen, 1944). According to Schmidt-Nielsen, this may explain the ability of the camel to maintain a constant blood volume. The half-life of T-1824 in normal *A. cahirinus* is shorter than in the rat and in *M. crassus*, either normal or dehydrated. We therefore cannot explain the ability of *A. cahirinus* to maintain its plasma volume by relating it to regular slow turnover of the plasma proteins. It seems more probable that, as a result of dehydration, changes occur in the permeability of blood capillaries to proteins. Work in progress supports these findings. The relative small reduction of PV in *M. crassus* is probably due to its inability to survive large losses of body water, and not because of any special mechanism of PV regulation. Its successful colonization of desert biotopes is due chiefly to small evaporation and behavioral adaptation—staying in protected burrows during the day time.

Of the three species tested under the extreme dehydration conditions, the white laboratory rat survived the largest fluid and blood plasma volume loss, a testimony to the hardiness of this species.

*M. crassus* could not sustain large fluid losses possibly because the entire burden of dehydration is borne by its extracellular fluid compartment.

*A. cahirinus* can survive larger losses of body water than *M. crassus* but less than the rat, although it can survive longer dehydration periods than the rat. It keeps almost a constant blood plasma volume, which is comparatively small, as well as a constant ECV. Most of its water loss is presumably from intracellular sources. The blood plasma volume conservation is probably connected with a decrease in the protein permeability of the blood capillaries.

The possible advantage of blood plasma volume conservation is the ability to survive over longer periods of dehydration.

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*Key Word Index*—*Rattus norvegicus*; *Acomys cahirinus*; *Meriones crassus*; desert rodents; dehydration; blood plasma volume; extracellular fluid volume; body water; capillary permeability.